

| Piezo ID | d_1 (mm)   | d_2 (mm)   | meas.with | resol. (mm) | t_1 (mm)    | t_2 (mm)    | t_3 (mm)    | meas.with  | resol. (mm) | C (nF)     | meas.with  | resol. (nF) | K          | Obs.                                                                            |
|----------|------------|------------|-----------|-------------|-------------|-------------|-------------|------------|-------------|------------|------------|-------------|------------|---------------------------------------------------------------------------------|
| 1        | 26.95      | 18.80      | caliper   | 0.05        | 0.326       | 0.133       | 0.193       | micrometer | 0.001       | 26.3       | multimeter | 0.1         | 2066       | Heavily stained piezo ceramic                                                   |
| 2        | 26.95      | 18.90      | caliper   | 0.05        | 0.327       | 0.132       | 0.195       | micrometer | 0.001       | 27.3       | multimeter | 0.1         | 2144       | Stained piezo ceramic                                                           |
| 3        | 26.95      | 19.10      | caliper   | 0.05        | 0.338       | 0.137       | 0.201       | micrometer | 0.001       | 28.4       | multimeter | 0.1         | 2251       | Stained piezo ceramic                                                           |
| 4        | 26.95      | 19.00      | caliper   | 0.05        | 0.370       | 0.155       | 0.215       | micrometer | 0.001       | 23.9       | multimeter | 0.1         | 2048       | Heavily stained piezo ceramic and brass w/ residual glue from previous assembly |
| 5        | 26.95      | 18.40      | caliper   | 0.05        | 0.335       | 0.131       | 0.204       | micrometer | 0.001       | 22.2       | multimeter | 0.1         | 1924       | -                                                                               |
| 6        | 26.95      | 19.00      | caliper   | 0.05        | 0.333       | 0.135       | 0.198       | micrometer | 0.001       | 26.4       | multimeter | 0.1         | 2083       | Stained piezo ceramic                                                           |
| 7        | 26.95      | 18.85      | caliper   | 0.05        | 0.337       | 0.135       | 0.202       | micrometer | 0.001       | 21.6       | multimeter | 0.1         | 1767       | Stained piezo ceramic                                                           |
| 8        | 26.95      | 19.00      | caliper   | 0.05        | 0.331       | 0.134       | 0.197       | micrometer | 0.001       | 27.2       | multimeter | 0.1         | 2135       | Stained piezo ceramic                                                           |
| 9        | 26.95      | 18.50      | caliper   | 0.05        | 0.359       | 0.133       | 0.226       | micrometer | 0.001       | 22.1       | multimeter | 0.1         | 2100       | Stained piezo ceramic                                                           |
| 10       | 26.95      | 18.90      | caliper   | 0.05        | 0.329       | 0.131       | 0.198       | micrometer | 0.001       | 25.2       | multimeter | 0.1         | 2010       | Heavily stained piezo ceramic                                                   |
| 11       | 26.95      | 18.85      | caliper   | 0.05        | 0.334       | 0.135       | 0.199       | micrometer | 0.001       | 27.5       | multimeter | 0.1         | 2216       | Stained piezo ceramic                                                           |
| 12       | 26.95      | 18.95      | caliper   | 0.05        | 0.354       | 0.155       | 0.199       | micrometer | 0.001       | 24.3       | multimeter | 0.1         | 1937       | Stained piezo ceramic and brass w/ residual glue from previous assembly         |
| 13       | 26.95      | 19.10      | caliper   | 0.05        | 0.339       | 0.136       | 0.203       | micrometer | 0.001       | 27.1       | multimeter | 0.1         | 2170       | -                                                                               |
| 14       | 26.95      | 19.00      | caliper   | 0.05        | 0.329       | 0.133       | 0.196       | micrometer | 0.001       | 26.5       | multimeter | 0.1         | 2070       | -                                                                               |
| 15       | 26.95      | 19.10      | caliper   | 0.05        | 0.332       | 0.137       | 0.195       | micrometer | 0.001       | 26.9       | multimeter | 0.1         | 2069       | -                                                                               |
| 16       | 26.95      | 19.05      | caliper   | 0.05        | 0.332       | 0.128       | 0.204       | micrometer | 0.001       | 25.5       | multimeter | 0.1         | 2062       | -                                                                               |
| 17       | 26.95      | 19.10      | caliper   | 0.05        | 0.327       | 0.128       | 0.199       | micrometer | 0.001       | 28.3       | multimeter | 0.1         | 2221       | Stained piezo ceramic                                                           |
| 18       | 26.95      | 19.20      | caliper   | 0.05        | 0.330       | 0.131       | 0.199       | micrometer | 0.001       | 26.7       | multimeter | 0.1         | 2074       | -                                                                               |
| 19       | 26.95      | 18.80      | caliper   | 0.05        | 0.340       | 0.130       | 0.210       | micrometer | 0.001       | 21.8       | multimeter | 0.1         | 1863       | Stained piezo ceramic                                                           |
| 20       | 26.95      | 18.75      | caliper   | 0.05        | 0.342       | 0.134       | 0.208       | micrometer | 0.001       | 21.3       | multimeter | 0.1         | 1813       | Stained piezo ceramic                                                           |
| 21       | 26.95      | 19.20      | caliper   | 0.05        | 0.328       | 0.131       | 0.197       | micrometer | 0.001       | 27.6       | multimeter | 0.1         | 2122       | -                                                                               |
| 22       | 26.95      | 18.80      | caliper   | 0.05        | 0.333       | 0.132       | 0.201       | micrometer | 0.001       | 23.8       | multimeter | 0.1         | 1947       | Stained piezo ceramic                                                           |
| 23       | 26.95      | 18.85      | caliper   | 0.05        | 0.324       | 0.131       | 0.193       | micrometer | 0.001       | 24.9       | multimeter | 0.1         | 1946       | -                                                                               |
| 24       | 26.95      | 19.00      | caliper   | 0.05        | 0.331       | 0.132       | 0.199       | micrometer | 0.001       | 26.6       | multimeter | 0.1         | 2110       | Stained piezo ceramic                                                           |
| 25       | 26.95      | 18.70      | caliper   | 0.05        | 0.339       | 0.132       | 0.207       | micrometer | 0.001       | 23.5       | multimeter | 0.1         | 2001       | -                                                                               |
| 26       | 26.95      | 19.10      | caliper   | 0.05        | 0.333       | 0.132       | 0.201       | micrometer | 0.001       | 26.5       | multimeter | 0.1         | 2101       | -                                                                               |
| 27       | 26.95      | 19.00      | caliper   | 0.05        | 0.334       | 0.134       | 0.200       | micrometer | 0.001       | 26.5       | multimeter | 0.1         | 2112       | -                                                                               |
| 28       | 26.95      | 18.55      | caliper   | 0.05        | 0.340       | 0.130       | 0.210       | micrometer | 0.001       | 21.4       | multimeter | 0.1         | 1879       | -                                                                               |
| 29       | 26.95      | 19.10      | caliper   | 0.05        | 0.332       | 0.131       | 0.201       | micrometer | 0.001       | 28.0       | multimeter | 0.1         | 2219       | -                                                                               |
| 30       | 26.95      | 18.65      | caliper   | 0.05        | 0.334       | 0.130       | 0.204       | micrometer | 0.001       | 23.0       | multimeter | 0.1         | 1941       | Stained piezo ceramic                                                           |
| 31       | 26.95      | 18.90      | caliper   | 0.05        | 0.320       | 0.126       | 0.194       | micrometer | 0.001       | 25.7       | multimeter | 0.1         | 2008       | -                                                                               |
| 32       | 26.95      | 18.90      | caliper   | 0.05        | 0.327       | 0.131       | 0.196       | micrometer | 0.001       | 26.8       | multimeter | 0.1         | 2116       | Stained piezo ceramic                                                           |
| 33       | 26.95      | 18.70      | caliper   | 0.05        | 0.337       | 0.134       | 0.203       | micrometer | 0.001       | 25.2       | multimeter | 0.1         | 2105       | -                                                                               |
| 34       | 26.95      | 19.00      | caliper   | 0.05        | 0.327       | 0.127       | 0.200       | micrometer | 0.001       | 26.1       | multimeter | 0.1         | 2080       | Stained piezo ceramic                                                           |
| 35       | 26.95      | 18.75      | caliper   | 0.05        | 0.338       | 0.131       | 0.207       | micrometer | 0.001       | 23.9       | multimeter | 0.1         | 2025       | Stained piezo ceramic                                                           |
| 36       | 26.95      | 18.45      | caliper   | 0.05        | 0.339       | 0.132       | 0.207       | micrometer | 0.001       | 21.0       | multimeter | 0.1         | 1837       | Stained piezo ceramic                                                           |
| 37       | 26.95      | 19.10      | caliper   | 0.05        | 0.333       | 0.133       | 0.200       | micrometer | 0.001       | 27.1       | multimeter | 0.1         | 2137       | Stained piezo ceramic                                                           |
| 38       | 27.00      | 19.10      | caliper   | 0.05        | 0.330       | 0.128       | 0.202       | micrometer | 0.001       | 27.2       | multimeter | 0.1         | 2167       | Stained piezo ceramic                                                           |
| 39       | 26.95      | 19.05      | caliper   | 0.05        | 0.330       | 0.133       | 0.197       | micrometer | 0.001       | 28.2       | multimeter | 0.1         | 2202       | -                                                                               |
| 40       | 26.95      | 18.70      | caliper   | 0.05        | 0.338       | 0.132       | 0.206       | micrometer | 0.001       | 23.6       | multimeter | 0.1         | 2000       | Stained piezo ceramic                                                           |
| 41       | 26.95      | 18.70      | caliper   | 0.05        | 0.343       | 0.134       | 0.209       | micrometer | 0.001       | 24.6       | multimeter | 0.1         | 2115       | Stained piezo ceramic                                                           |
| 42       | 26.95      | 19.10      | caliper   | 0.05        | 0.332       | 0.131       | 0.201       | micrometer | 0.001       | 27.6       | multimeter | 0.1         | 2188       | Stained piezo ceramic                                                           |
| 43       | 26.95      | 19.10      | caliper   | 0.05        | 0.328       | 0.134       | 0.194       | micrometer | 0.001       | 26.5       | multimeter | 0.1         | 2027       | Stained piezo ceramic                                                           |
| 44       | 26.95      | 18.55      | caliper   | 0.05        | 0.335       | 0.130       | 0.205       | micrometer | 0.001       | 23.0       | multimeter | 0.1         | 1971       | -                                                                               |
| 45       | 26.95      | 18.90      | caliper   | 0.05        | 0.327       | 0.131       | 0.196       | micrometer | 0.001       | 26.7       | multimeter | 0.1         | 2108       | Stained piezo ceramic                                                           |
| 46       | 26.95      | 18.75      | caliper   | 0.05        | 0.339       | 0.133       | 0.206       | micrometer | 0.001       | 20.7       | multimeter | 0.1         | 1745       | Stained piezo ceramic                                                           |
| 47       | 26.95      | 19.00      | caliper   | 0.05        | 0.320       | 0.128       | 0.192       | micrometer | 0.001       | 27.6       | multimeter | 0.1         | 2112       | -                                                                               |
| 48       | 26.95      | 18.75      | caliper   | 0.05        | 0.338       | 0.132       | 0.206       | micrometer | 0.001       | 23.6       | multimeter | 0.1         | 1989       | Stained piezo ceramic                                                           |
| 49       | 26.95      | 18.70      | caliper   | 0.05        | 0.353       | 0.141       | 0.212       | micrometer | 0.001       | 24.4       | multimeter | 0.1         | 2128       | Stained piezo ceramic and brass w/ residual glue from previous assembly         |
| 50       | 26.95      | 19.00      | caliper   | 0.05        | 0.328       | 0.128       | 0.200       | micrometer | 0.001       | 24.0       | multimeter | 0.1         | 1913       | Stained piezo ceramic                                                           |
| Average  | 27.0 ± 0.0 | 18.9 ± 0.2 | -         | -           | 0.34 ± 0.01 | 0.13 ± 0.01 | 0.20 ± 0.01 | -          | -           | 25.2 ± 2.2 | -          | -           | 2048 ± 119 | -                                                                               |
| Total    | -          | -          | -         | -           | -           | -           | -           | -          | -           | 1261.8     | -          | -           | -          | -                                                                               |

| Legend |                                            |
|--------|--------------------------------------------|
| d_1    | Diameter of the brass disk                 |
| d_2    | Diameter of the piezoelectric ceramic disk |
| t_1    | Total thickness                            |
| t_2    | Thickness of the brass layer               |
| t_3    | Thickness of the piezoelectric layer       |
| C      | Capacitance of the piezoelectric disk      |
| K      | Dielectric contant of the piezo ceramic    |